

What breaks ESP32 DMX receive: an interrupt stall past the FIFO budget

Deliberately blocking interrupts for 8 ms ($>$ the 5.6 ms FIFO) corrupts receive ($\sim 13\times$ baseline).
Real flash writes during receive do NOT: esp_dmx keeps its ISR in IRAM and runs through the cache
blackout. A DMX library WITHOUT an IRAM ISR would corrupt here exactly like the blocked-ISR case.

framing / overrun errors (60 s run)

2 errors
in 2196 frames

baseline
(no WiFi)

25 errors
in 2108 frames

interrupts blocked
8 ms $>$ 5.6 ms FIFO

3 errors
in 2059 frames

flash writes
during receive
(esp_dmx IRAM ISR)

